

DATABASE SYSTEMS

CHAP. 1: Overview

Database

- Integrated collection of stored operational data used by the application systems of a particular enterprise.
- Example:
 - . Mass storage-weather records
 - . Telephone records
 - . Electoral rolls

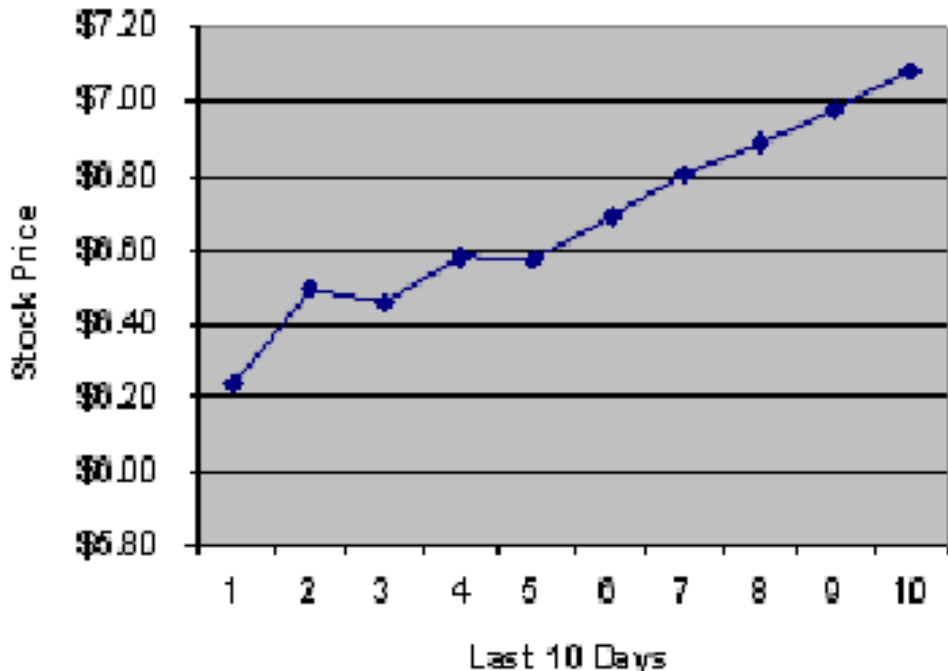
Data

- Representations of raw facts, concepts, or instructions in a formalized manner suitable for communication, interpretation, or processing by humans or by automatic means.
- Any representation such as characters or analog quantities to which meaning is or might be assigned.

Information

- Meaning that a human assigns to data by means of the known conventions used in their representation.

Data & Information

Data	Information																						
6.34	SIRIUS SATELLITE RADIO INC.  <table><caption>Stock Price Data for Sirius Satellite Radio Inc.</caption><tr><th>Day</th><th>Stock Price</th></tr><tr><td>1</td><td>\$6.25</td></tr><tr><td>2</td><td>\$6.55</td></tr><tr><td>3</td><td>\$6.45</td></tr><tr><td>4</td><td>\$6.55</td></tr><tr><td>5</td><td>\$6.55</td></tr><tr><td>6</td><td>\$6.65</td></tr><tr><td>7</td><td>\$6.80</td></tr><tr><td>8</td><td>\$6.90</td></tr><tr><td>9</td><td>\$7.00</td></tr><tr><td>10</td><td>\$7.10</td></tr></table>	Day	Stock Price	1	\$6.25	2	\$6.55	3	\$6.45	4	\$6.55	5	\$6.55	6	\$6.65	7	\$6.80	8	\$6.90	9	\$7.00	10	\$7.10
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Field, Records, Files

- **Field (Columns)**

- Group of characters with specific meaning, headings corresponding to the attributes of the object.

- **Record (Row)**

- Logically connected fields that describe a person, place, or thing, one set of attributes (or one tuple) corresponding to one instance of the entity that a table schema describes.

- **File**

- Collection of related records

DBMS

- **Database Management System (DBMS)**

- Collection of programs that enables users to create and maintain the database.

Advantages of DBMS

- Controlling redundancy
- Providing storage structure for efficient query processing.
- Restricting unauthorized users.
- Providing concurrency.
- Providing backup and recovery.
- Enforcing integrity constraints.

Disadvantages of DBMS

- Centralization: That is use of the same program at a time by many user sometimes lead to loss of some data.
- High cost of software.
- Technical expertises are required
- Power dependency
- Reporting features like charts of a spreadsheet like Excel may not be available in RDBMS.

Data model

- Set of concepts that can be used to describe the structure of a database: the data types, relationships, constraints, semantics and operational behaviour.
- It is a tool for data abstraction.

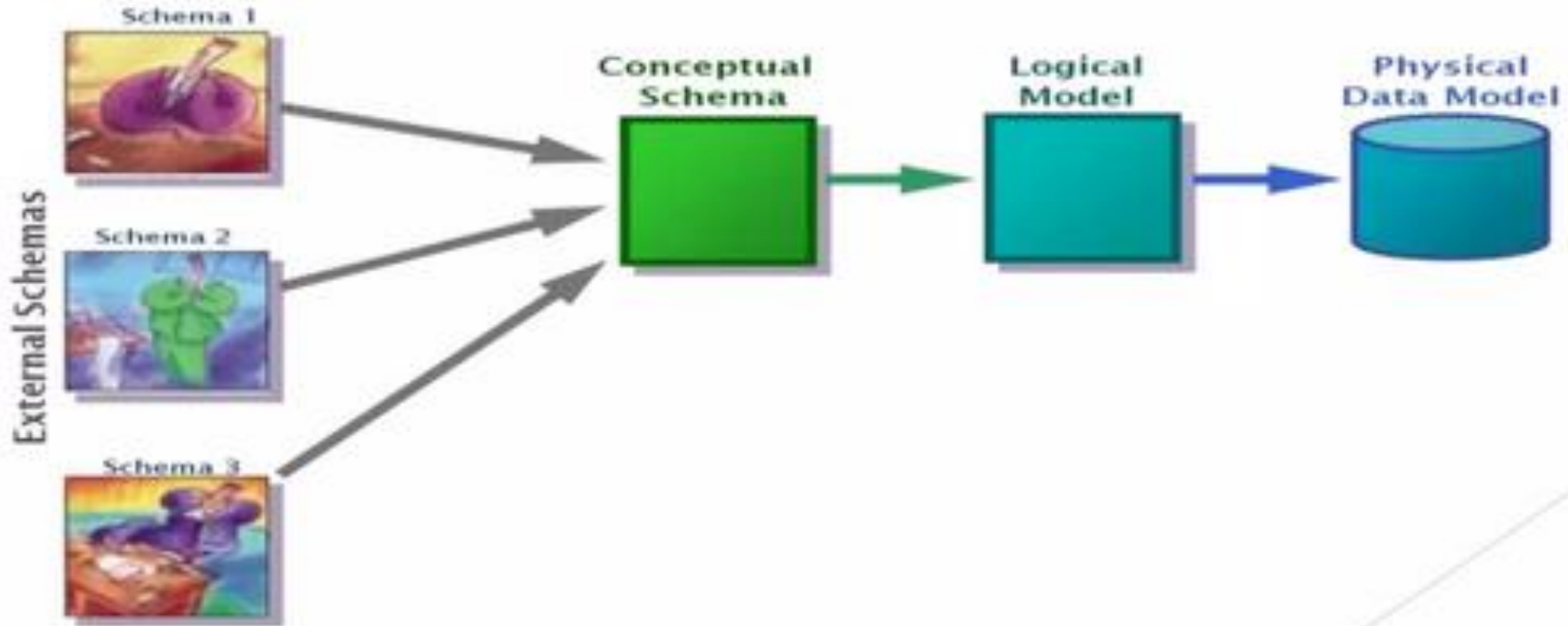
Data model operations

- Operations for specifying database retrievals and updates by referring to the concepts of the data model.
- Operations on the data model may include basic operations and user-defined operations.

Categories of data models

- **Conceptual:**
 - Describes WHAT the system contains.
- **Logical:**
 - Describes how the system will be implemented, regardless of the DBMS
- **Physical:**
 - Describes HOW the system will be implemented using a specific DBMS.

Categories of data models



Database Models

- Theoretical foundation of a database and fundamentally determines in which manner data can be stored, organized, and manipulated in a database system.
- It thereby defines the infrastructure offered by a particular database system.

Database Models

- Flat
- Hierarchical
- Network
- Relational
 - Object-Oriented
 - Object-Relational
 - NoSQL
 - XML

Flat Database Model

- Aka table model, consists of a single, two-dimensional array of data elements, where all members of a given column are assumed to be similar values, and all members of a row are assumed to be related to one another.

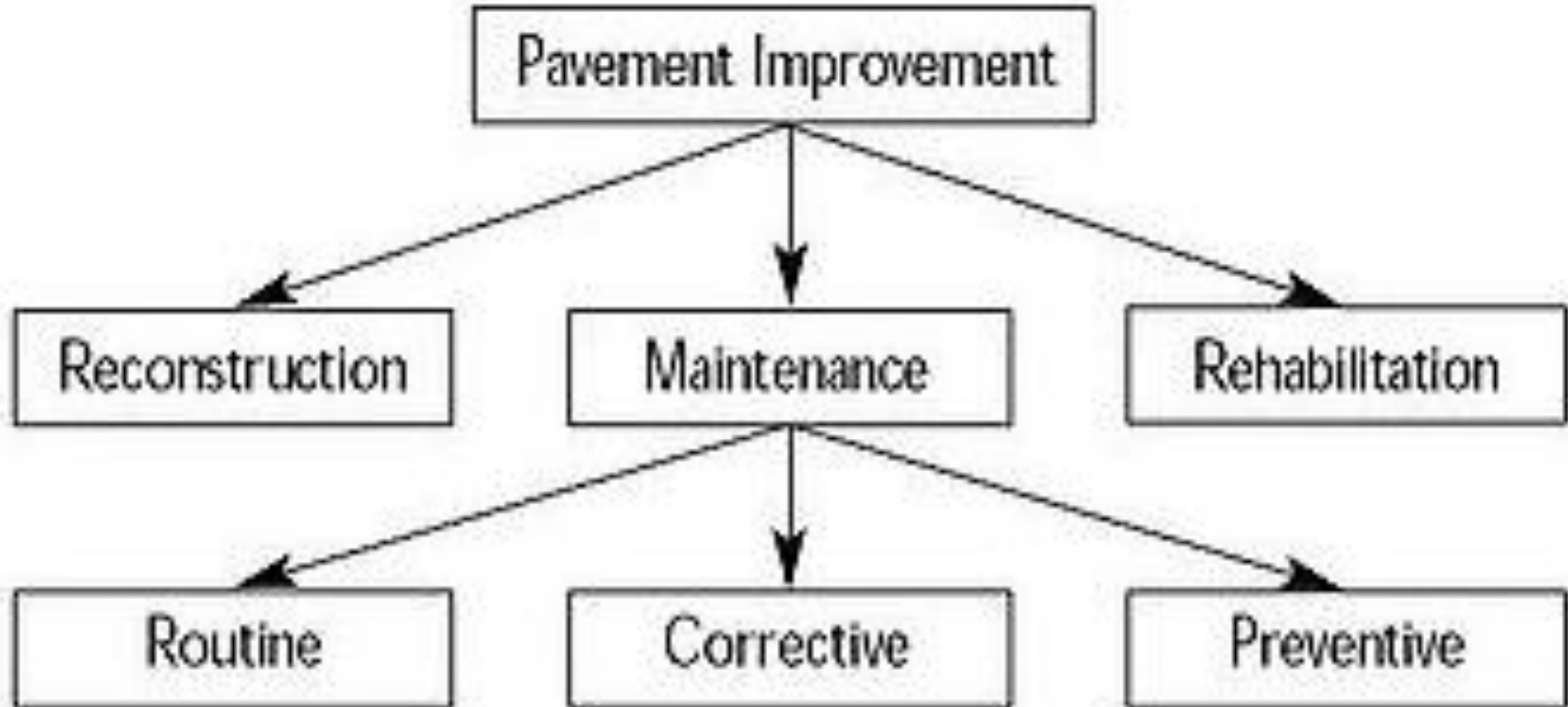
Flat Database Model

	Route No.	Miles	Activity
Record 1	I-95	12	Overlay
Record 2	I-495	05	Patching
Record 3	SR-301	33	Crack seal

Hierarchical Database Model

- Data in a tree structure .
- Hierarchy of parent and child data segments, this structure implies that a record can have repeating information, generally in the child data segments, data in a series of records, which have a set of field values attached to it

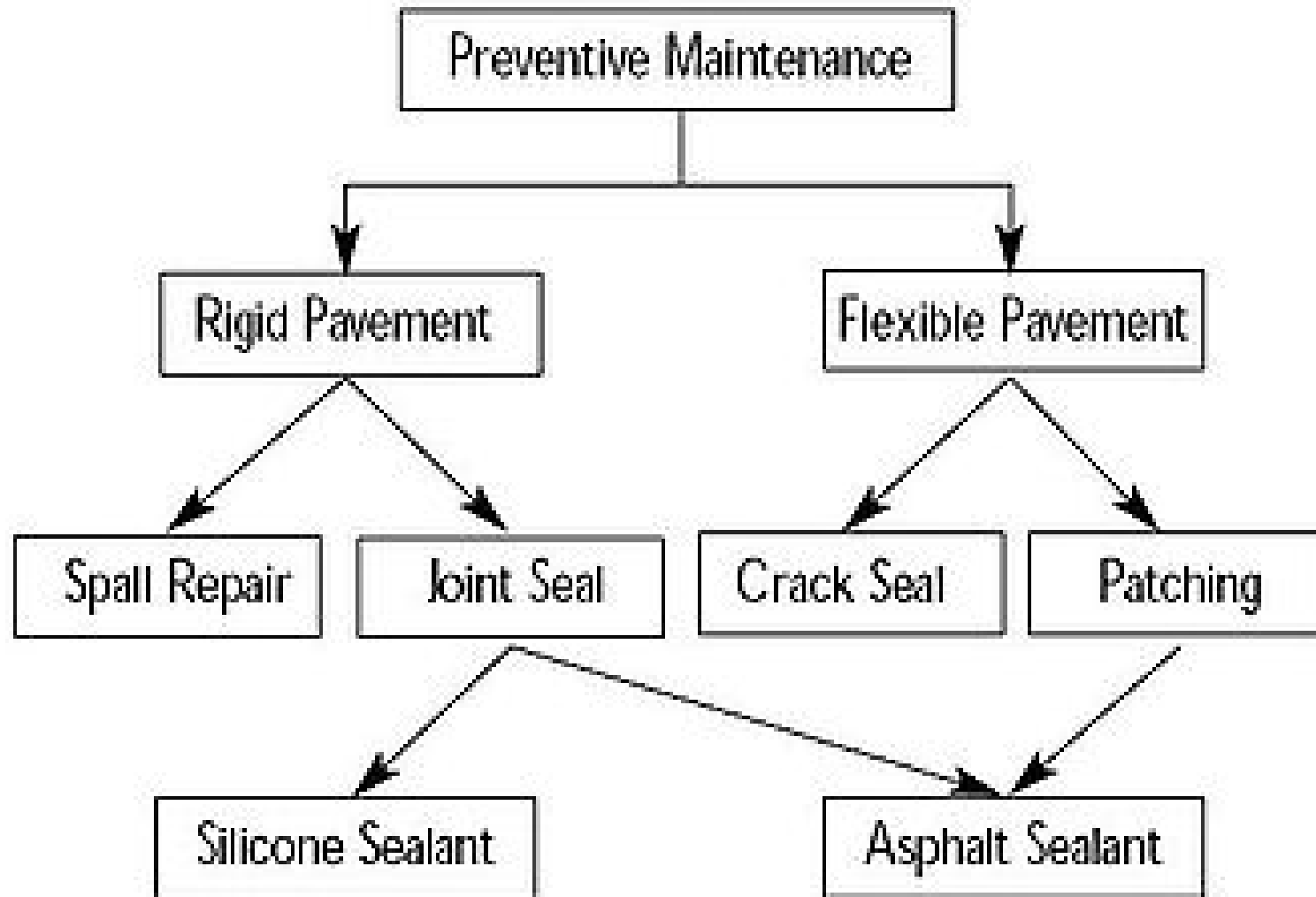
Hierarchical Database Model



Network Database Model

- Organizes data using two fundamental constructs, called records and sets.
- Records contain fields
- Sets define one-to-many relationships between records: one owner, many members.
- A record may be an owner in any number of sets, and a member in any number of sets.

Network Database Model



Relational Database Model

- Allows the definition of data structures, storage and retrieval operations and integrity constraints.
- Data and relations between them are organized in tables.
- Table is a collection of records and each record in a table contains the same fields.

Relational Database Model

Activity Code	Activity Name
23	Patching
24	Overlay
25	Crack Sealing

Key = 24

Activity Code	Date	Route No.
24	01/12/01	I-95
24	02/08/01	I-66

Date	Activity Code	Route No.
01/12/01	24	I-95
01/15/01	23	I-495
02/08/01	24	I-66

Object-Oriented Database Model

- Based on object-oriented programming, associating methods with objects that can benefit from class hierarchies.
- Thus, “objects” are levels of abstraction that include attributes and behavior.

Object- Relational Database Model

- Combination of a Object oriented database model and a Relational database model.
- Supports objects, classes, inheritance etc. just like Object Oriented models and has support for data types, tabular structures etc. like Relational data model.

NoSQL Database Model

- Model that is not reinforced by a relational database management system (RDBMS).
- Model isn't explicit about how the data relates – how it all connects together.

XML Database Model

- Described in terms of sequences and items, atomic values, and nodes.
- The XPath data model is based on the notion of a sequence.
- The value of an XPath expression is always a sequence.
- A sequence is an ordered collection of zero or more items.

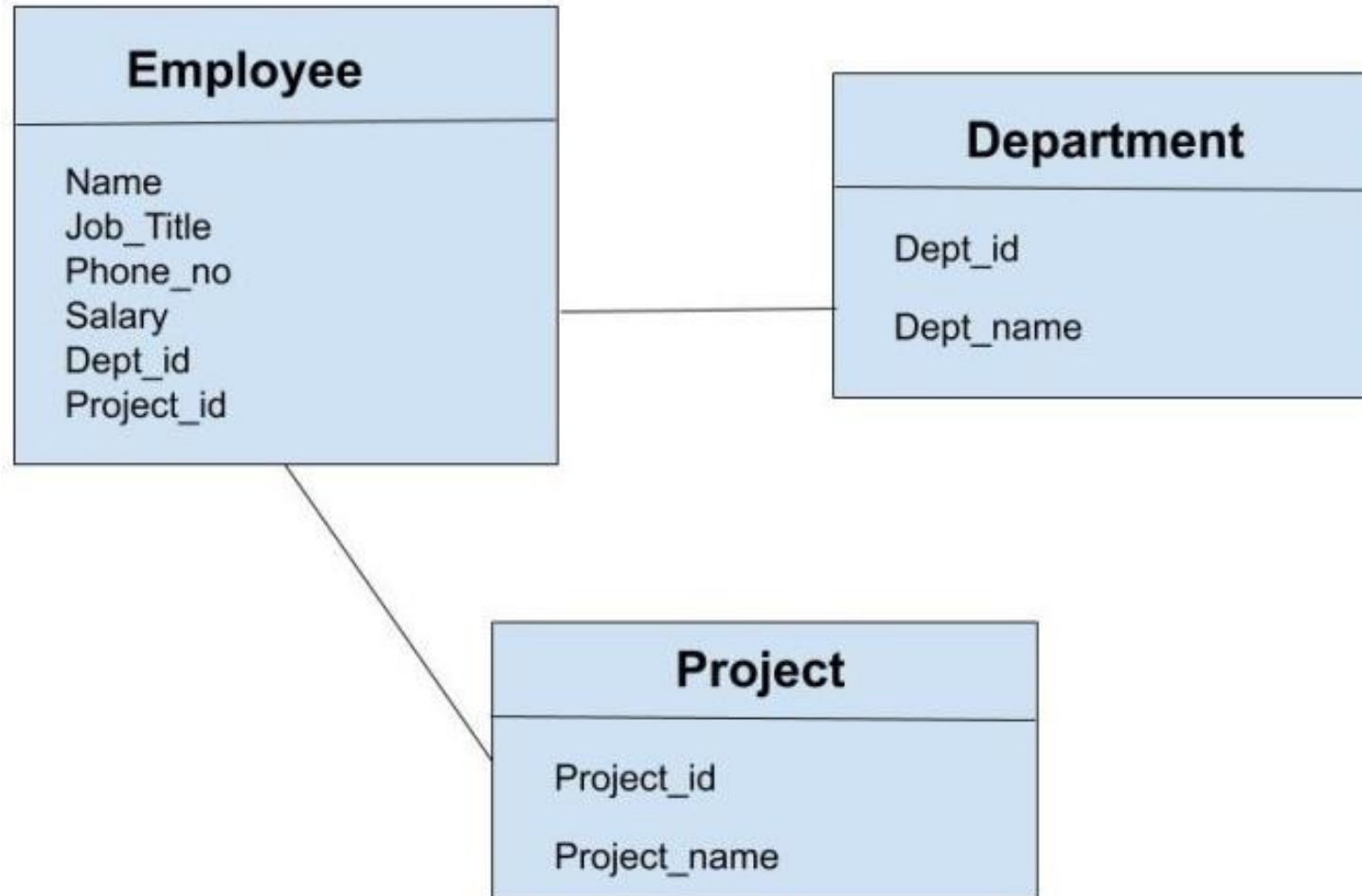
Database Schema

- Structure that represents the logical storage of the data in a database.
- It represents the organization of data and provides information about the relationships between the tables in a given database.

Schema Diagram

- Diagrammatic display of (some aspects of) a database schema.

Schema Diagram



Database Architecture

- Set of specifications, rules, and processes that dictate how data is stored in a database and how data is accessed by components of a system.
- It includes data types, relationships, and naming conventions.

Database Architecture (Cont.)

- Describes the organization of all database objects and how they work together.
- It affects integrity, reliability, scalability, and performance.
- Involves anything that defines the nature of the data, the structure of the data, or how the data flows.

Three Schemas Architecture

- External schema (many schemas may exist)
- Conceptual schema (only one per database)
- Internal schema (only one per database)

External Schema

- Describes a subset of the database that a particular user group is interested in, according to the format user wants, and hides the rest
- May contain virtual data that is derived from the files, but is not explicitly stored

Conceptual Schema

- Hides the details of physical storage structures and concentrates on describing entities, data types, relationships, operations, and constraints.

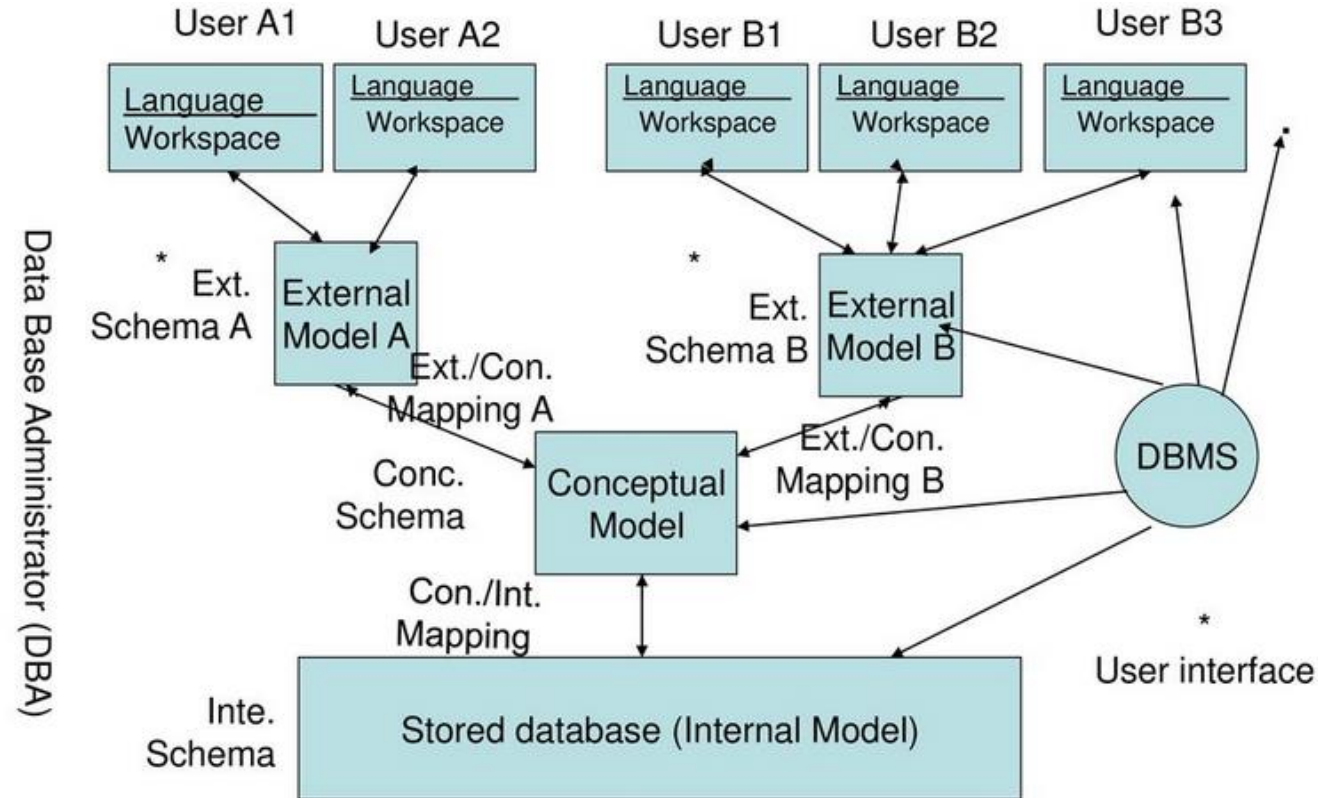
Internal Schema

- Describes the physical storage structure of the DB - uses a low-level (physical) data model to describe the complete details of data storage and access paths

Three Schemas Architecture(Cont.)

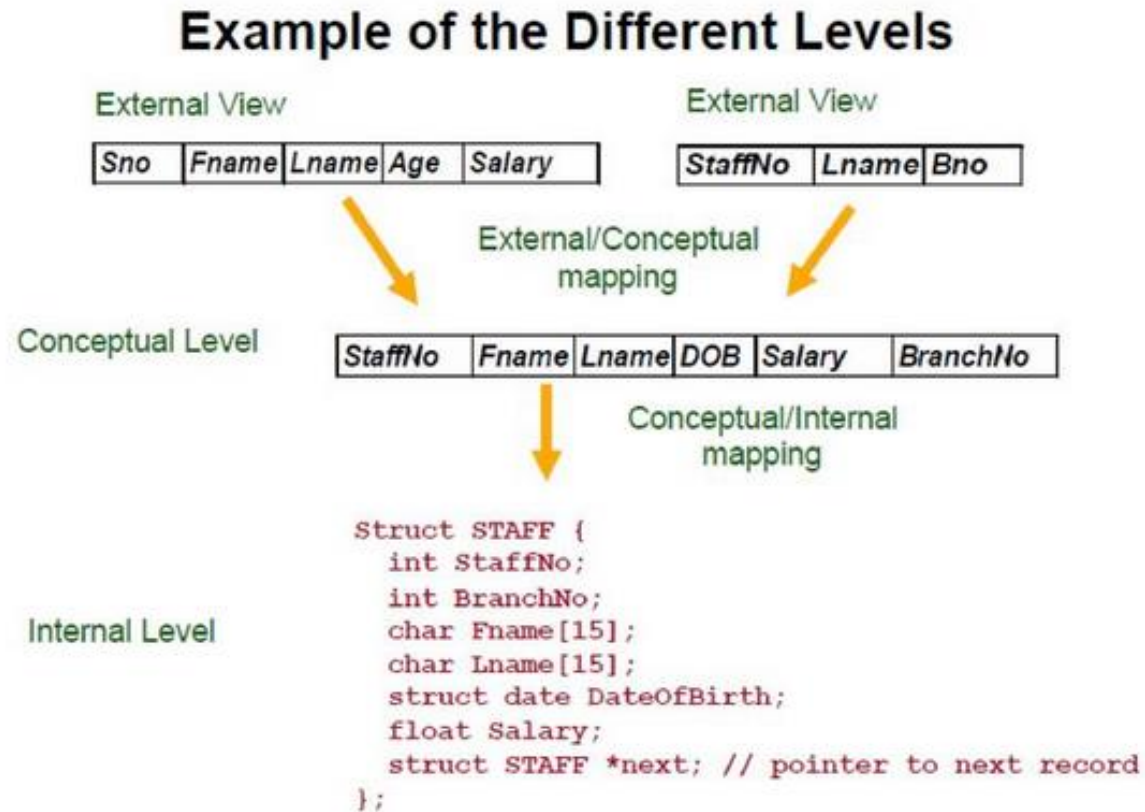
- The DBMS is responsible for mapping between the three types of schema (i.e. how they actually correspond with each other).
- Each external schema is related to the conceptual schema (external/conceptual mapping).
- A conceptual/internal mapping relates the conceptual schema to the internal schema.

Three Schemas Architecture(Cont.)



An Architecture for Database System

Three Schemas Architecture(Cont.)



Database languages

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Data control language (DCL)
- Transaction control language (TCL)

DDL

- Creates the framework of the database by specifying the database schema, which is the structure that represents the organization of data.
- Its common uses include the creation and alteration of tables, files, indexes and columns within the database.
- This language also allows users to rename or drop the existing database or its components.

DDL

- . CREATE: Creates a new database or object, such as a table, index or column
- . ALTER: Changes the structure of the database or object
- . DROP: Deletes the database or existing objects
- . RENAME: Renames the database or existing objects

DML

- . Provides operations that handle user requests, offering a way to access and manipulate the data that users store within a database.
- . Common functions include inserting, updating and retrieving data from the database.

DML

- INSERT: Adds new data to the existing database table
- UPDATE: Changes or updates values in the table
- DELETE: Removes records or rows from the table
- SELECT: Retrieves data from the table or multiple tables

DCL

- Controls access to the data that users store within a database.
- Controls the rights and permissions of the database system.
- Allows users to grant or revoke privileges to the database.

DCL

- . GRANT: Gives a user access to the database
- . REVOKE: Removes a user's access to the database

TCL

- . Manages the transactions within a database.
- . Transactions group a set of related tasks into a single, executable task.
- . All the tasks must succeed in order for the transaction to work.

TCL

- . COMMIT: Carries out a transaction
- . ROLLBACK: Restores a transaction if any tasks fail to execute
- . SAVEPOINT: Sets a point in a transaction to save
- . SET TRANSACTION: Places a name on a transaction.